testers that are specifically designed for the sub-systems/systems. Typically, a fully-assembled product testing involves creating the exact use-case situation with the tester and tests the product for its functionality, performance, and its fault-handling capability. Most of the ATEs (automatic test equipment) designed for this testing are hundred percent custom designed. The only common thing is that they may be built using platforms like LabView software along with compatible hardware modules.

With the emergence of scripting languages like Python, developers use open source software and custom hardware to build testers to keep the costs down. One lacuna seen with startups and SMEs is not adding a few hooks in their design, which leads to an expensive tester. A good example of test hook is the calibration signal output connector provided in most of the oscilloscopes, which enables the user to check by just connecting the probe to the signal generator (square wave of 1kHz @ 1V or 5V) pin provided for self-checking.

Except for software platforms like LabView or HP Vee, ATEs will be fully customised to suit the product that is under test. In the case of ICT, the tester and the fixture base remain the same except the BoN fixture and the ICT software which changes depending on the PCBA tested. Fig. 9 shows a typical view of an ATE.

ATEs are used for both sub-systems as well as finished systems. One of the key aspects most designers miss is that with a little design effort, the testing of the PCBs or systems can be easily implemented in a simple way at a lower cost.


Fig. 9: A typical view of an ATE

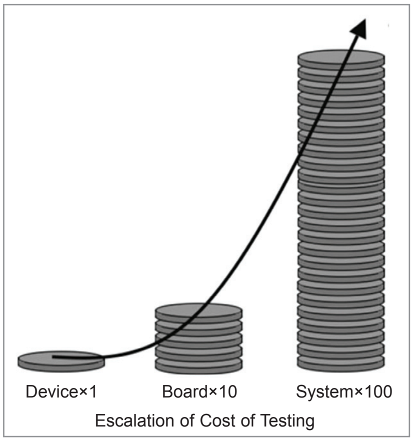

Fig. 10: Rule of ten

Standard compliance testing

With increased globalisation, embedded systems are finding markets globally. However, each country or continent tends to have specific standards for electrical safety and electrical emissions and susceptibility. Products have to be designed to meet these standards before they can be commercially sold in the respective regions.

These tests are typically done by specific certified test labs. However, the onus of the product meeting the requirement is on the designers as well as the manufacturing partner.

While these tests are typically carried out one-time with the products, regulatory agencies like FCC, CE and VCCI periodically get sample products from the market and test them for compliance to the standards. Any failure to meet these tests will result in these agencies suspending the product sale till the problems are rectified. This forces the designer to design the product with sufficient margins to address the process variations in the manufacturing line.

How testing impacts products

The quality of testing impacts embedded systems in multiple ways. The 'rule of ten' explains this impact graphically. This rule is derived based on historical data and is shown in Fig. 10.

We can see that if a component failure is detected early on, the cost is just the failed component and test charges related to the test. When the same component failure is detected in assembled board, the cost of testing goes up ten times (component cost + assembly cost + ICT test cost). And if the same problem is detected in the system, either in the manufacturing line or in the field, the cost is one hundred times.

This shows the importance of testing and complete testing of a product helps the product maintain quality at a low price. An insufficiently tested product results in:

- Product's failure in the customer's place, leading to customer dissatisfaction.
- Increased product support costs and repair costs.
- Company's brand gets damaged due to failures.
- This issue gets aggravated when the product is from a start-up or SME who is in the initial phases and may not have a big support network. Any field failure can result in customer dissatisfaction. As a result, most pilots do not go to the next stage and the product is deemed as a failure.
- Insufficiently tested product results in reduced sales and damaged reputation, something small companies and start-ups cannot afford. They should not rush to market with a poorly-tested product due to pressure. The key here is that designers plan and implement test hooks which increase the testability of the product and reduce failure with simpler testers.

Conclusion

Here, we have seen the different tests that an embedded system has to undergo and the different types of tests that are needed for the same. We also saw how the testing of embedded systems impacts product quality and functioning.

The primary motive of this article is to sensitise both hardware and software designers about the importance of designing testable products and how to enable low cost of testing. Unfortunately, academic courses do not teach these as the courses are focused on learning the fundamentals. Further, academic courses cover processors and controllers that are no longer in production, leading to a gap.

In my next article, we shall discuss how to implement testability in the design of embedded systems. EFY

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